Mathematical Preliminaries

Theory of Computation and Automata

1

- Mathematical Preliminaries ✓
- Sets ✓
- Functions ✓
- Relations ✓
- Graphs ✓
- Proof Techniques

2

SETS

A set is a collection of elements

$$A = \{1, 2, 3\}$$

$$B = \{train, bus, bicycle, airplane\}$$

We write

$$1 \in A$$

$$ship \notin B$$

Set Representations

finite
infinite

C = { a, b, c, d, e, f, g, h, i, j, k }

C = { a, b, ..., k } ⟶ *finite set*

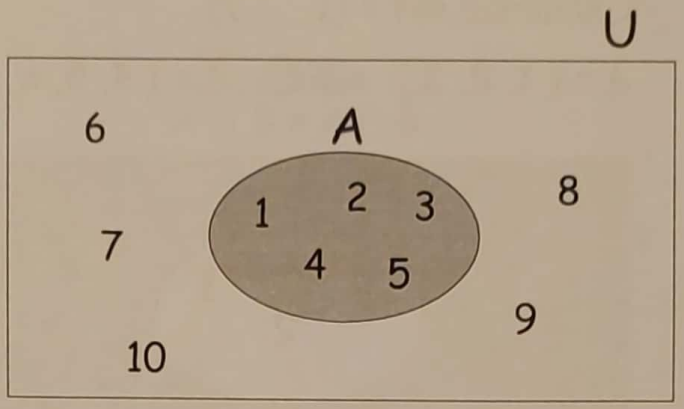
S_1 = { 2, 4, 6, ... } ⟶ *infinite set*

S_2 = {1, 3, 5, ...}

S_1 = { j : j > 0, and j = 2k for some k>0 }

S_1 = { j : j is nonnegative and even }

$A = \{1, 2, 3, 4, 5\}$



Universal Set: all possible elements

$U = \{1, \dots, 10\}$

* $A \subset U$
$\overline{U} = \Phi$
$\overline{U} = \{\}$

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5

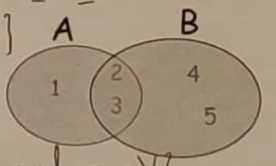
universal هي subset في set أي *

Set Operations

$A = \{1, 2, 3\}$ $B = \{2, 3, 4, 5\}$

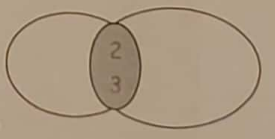
① • Union $X \cup Y = \{z \mid z \in X \text{ or } z \in Y\}$

$A \cup B = \{1, 2, 3, 4, 5\}$



② • Intersection $X \cap Y = \{z \mid z \in X \text{ and } z \in Y\}$

$A \cap B = \{2, 3\}$



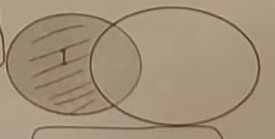
③ • Difference $X - Y = \{z \mid z \in X \text{ and } z \notin Y\}$

$A - B = \{1\}$

$B - A = \{4, 5\}$

$A - B \neq B - A$

$X - Y = X \cap \overline{Y}$



Venn diagrams

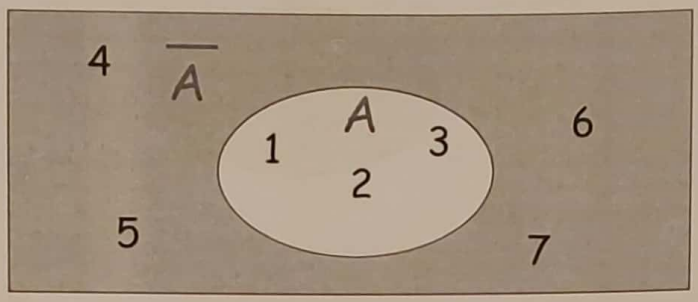
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6

- Complement

Universal set = {1, ..., 7}

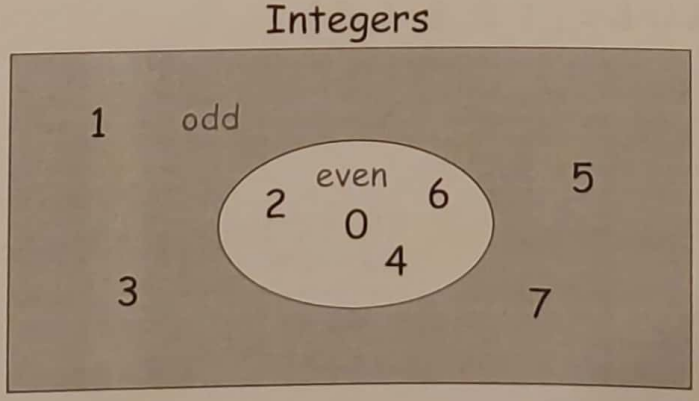
$A = \{1, 2, 3\}$ ⇒ $\overline{A} = \{4, 5, 6, 7\}$



$\overline{\overline{A}} = A$

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$\overline{\{\text{even integers}\}} = \{\text{odd integers}\}$

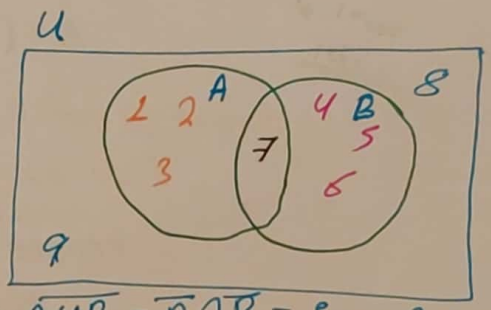


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DeMorgan's Laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$



* $\overline{A \cup B} = \overline{A} \cap \overline{B} = \{8, 9\}$
* $\bar{A} = \{4, 5, 6, 8, 9\}$
* $\bar{B} = \{1, 2, 3, 8, 9\}$
* $\bar{A} \cap \bar{B} = \{8, 9\}$

* $\bar{A} \cup \bar{B} = \{1, 2, 3, 4, 5, 6, 8, 9\}$

Empty, Null Set: $\emptyset$

$\emptyset = \{\}$

$S \cup \emptyset = S$

$S \cap \emptyset = \emptyset$

$S - \emptyset = S$

$\emptyset - S = \emptyset$

$\overline{\emptyset}$ = Universal Set

$\overline{U} = \emptyset$

Subset

* اعرف الفرق

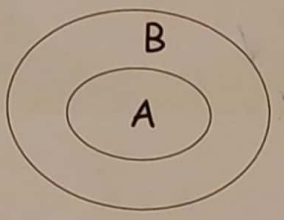
$A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$

$A \subseteq B$ ← ممكن A تساوي B وممكن A جزء من B

Proper Subset: $A \subset B$

A بدو يكون اقل بـ 1 ≡

A جزء من B بس مستحيل تكون A تساوي B



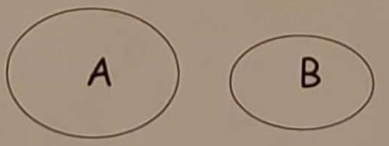
if $A \subset B$

1) $A \cap B = A$
2) $A \cup B = B$
3) $A - B = \emptyset$
4) $B - A = \{4, 5\}$

Disjoint Sets

$A = \{1, 2, 3\}$ $B = \{5, 6\}$

$A \cap B = \emptyset$ ← then A and B are disjoint sets.



Set Cardinality

- For finite sets

$A = \{ 2, 5, 7 \}$

$|A| = 3$

(set size)

$B = \{0, 1, 2, 5\}$

$|B| = 4 \qquad |\emptyset| = 0$

$|\{\}| = 0$

$C = \{ \{1\}, \{1,2\}, \{1,2,3\} \}$

$|C| = 3$

Powersets → P(S) or 2^S

A powerset is a set of sets

$S = \{ a, b, c \}$

$\{b, c\} \in P(S)$, $\emptyset \in P(S)$

$S \in P(S)$

* كل الـ set إلي بقدر اتكلم عن S

Powerset of S = the set of all the subsets of S

* انتبه ما كتبنا aa لانو بالـ set ما بجوز التكرار.

$P(S) = 2^S = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$

$* 2^n$ = size of $P(S)$, where $n = |S|$

تعني انو عندي 5 احتمالات 0, 1, 2, 3, 4

Observation: $| 2^S | = 2^{|S|}$ $(8 = 2^3)$

$\emptyset$ ←

$S = \{a, b, c, d\} \qquad n = |S| = 4$

$P(S) = 2^S = \{ \overbrace{\emptyset}^{0}, \overbrace{\{a\}, \{B\}, \{c\}, \{d\}}^{1}, \underbrace{\{a,b\}, \{a,c\}, \{a,B\} \ldots}_{2} \}$

R

ما حطينا $\{b, b\}$ ← لانو ما بصير نكرر العناصر

ولا رجعنا كتبنا الـ $\{b, a\}$ لانها موجودة ← $\{a, B\}$.

Cartesian Product

$A = \{2, 4\}$ $B = \{2, 3, 5\}$

$A \times B = \{(2, 2), (2, 3), (2, 5), (4, 2), (4, 3), (4, 5)\}$

one element

$(2,3) \neq (3,2)$

$\{2,3\} = \{3,2\}$

$|A \times B| = |A| \cdot |B|$

$= 2 * 3 = 6$

ex :-

Generalizes to more than two sets $A = \{a, b\}$ $B = \{a, b, c\}$

$A \times B \times \ldots \times Z$

* $A \times B = \{(a,a), (a,b), (a,c)$
$(b,a), (b,b), (b,c)\}$

$B \times A = \{(a,a), (a,b), (b,a)$
$(b,b), (c,a), (c,b)\}$

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15

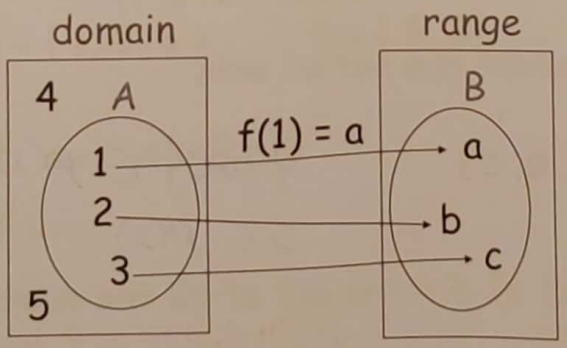
$A \times B \times A = \{(a,a,a), (a,a,b) \ldots$
$\ldots (b,c,a), (b,c,b)\}$ ☹ $A \times B \neq B \times A$

موجودة

∴ بنوخذ اول عنصر من A×B , بعدين A بـ cartesian

* $|A \times B \times A| = |A| \cdot |B| \cdot |A| = 2 * 3 * 2 = 12.$

FUNCTIONS



Partial function. → domain ≠ A

$f : A \rightarrow B$

If A = domain

then f is a total function

otherwise f is a partial function

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16

RELATIONS

$R = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots\}$

$x_i \, R \, y_i$

e. g. if R = '>': $2 > 1$, $3 > 2$, $3 > 1$

Equivalence Relations

- Reflexive: $x \, R \, x$
- Symmetric: $x \, R \, y \Rightarrow y \, R \, x$
- Transitive: $x \, R \, y$ and $y \, R \, z \Rightarrow x \, R \, z$

Example: R = '='

- $x = x$
- $x = y \Rightarrow y = x$
- $x = y$ and $y = z \Rightarrow x = z$

Equivalence Classes

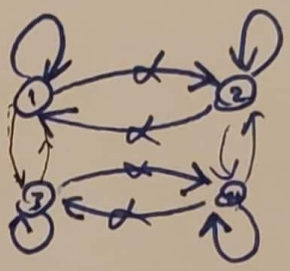
For equivalence relation R

equivalence class of x = {y : x R y}

شرح الدكتور *

Example:

R = { (1, 1), (2, 2), (1, 2), (2, 1),
(3, 3), (4, 4), (3, 4), (4, 3) }



Equivalence class of 1 = {1, 2}

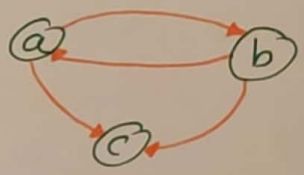
Equivalence class of 3 = {3, 4}

GRAPHS

G=(V,E)

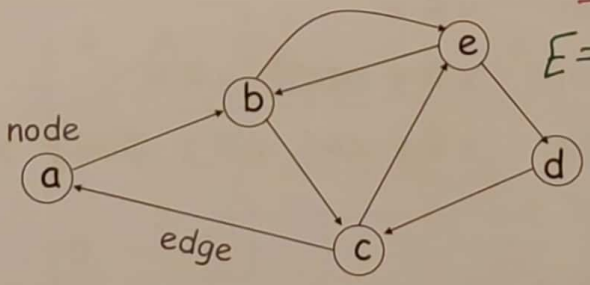
E ⊆ V×V

V = {a,b,c}



V×V = {(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,a), (c,b), (c,c)}

A directed graph



E = {(a,b), (b,a), (b,c), (a

E ⊆ V × V

لا تو ممكن

يكون كل V×V موجوان edge. ك

- Nodes (Vertices)

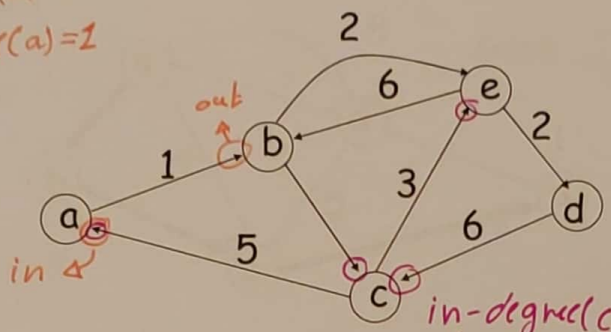
V = { a, b, c, d, e }

- Edges

E = { (a,b), (b,c), (b,e), (c,a), (c,e), (d,c), (e,b), (e,d) }

Labeled Graph

in-degree(a) = 1
out-degree(a) = 1



in-degree(c) = 2
out-degree(c) = 2

21

* simple ~~path~~ node

إذا شفتها node بتكون edge. مش

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contains each edge once.

euler tour

walk

path

no edge repeated.

simple path

No node repeated.

cycle

only base node is repeated.

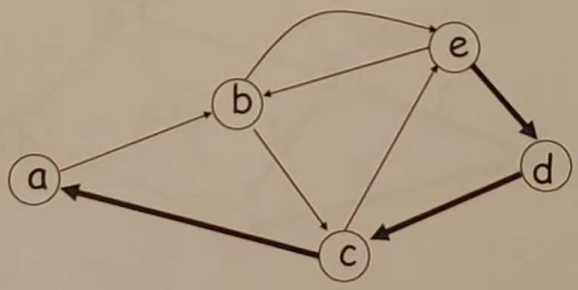
walk from base node to itself

simple cycle

Hamiltonian cycle

contains all nodes.

Walk



Walk is a sequence of adjacent edges

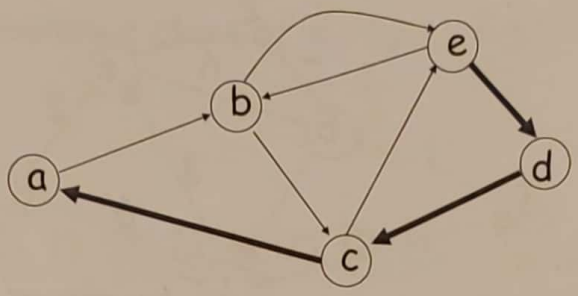
(e, d), (d, c), (c, a)

e → d → c → a

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22

Path

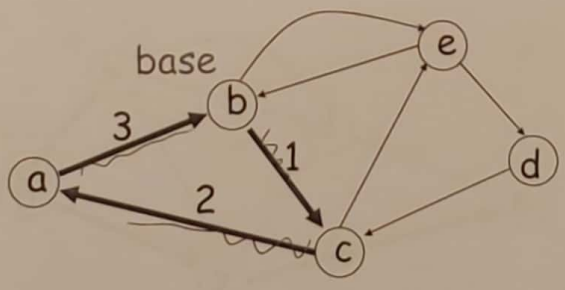


Path is a walk where no edge is repeated.

Simple path: no node is repeated

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Cycle



Cycle: a walk from a node (base) to itself

Simple cycle: only the base node is repeated

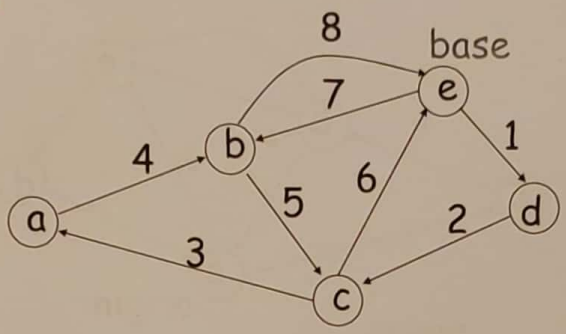
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Euler Tour

اعرف الفرق بينهما ← Euler Tour

simple cycle ← node

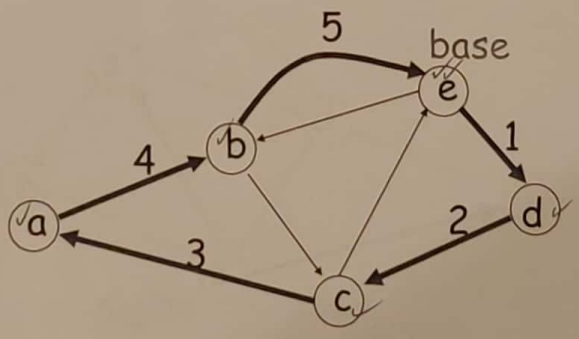
وش الـ euler tour ← edge.



A cycle that contains each edge once

e → d → c → a → b → c → e → b → e

Hamiltonian Cycle



* A simple cycle that contains all nodes.

Languages- Strings - Symbols

1

Language: *a set of strings*

String: *a sequence of symbols from some alphabet*

sigma (Σ)

Example:

Strings: cat, dog, house

Language: {cat, dog, house}

Alphabet: $\Sigma = \{a, b, c, \ldots, z\}$

2

Languages are used to describe computation problems:

$$PRIMES = \{2,3,5,7,11,13,17,\ldots\}$$

$$EVEN = \{0,2,4,6,\ldots\}$$

Alphabet: $\Sigma = \{0,1,2,\ldots,9\}$

Computation is translated to set membership
↳ $\in$

Example computation problem:

Is number x prime?

Equivalent set membership problem:

$x \in PRIMES = \{2,3,5,7,11,13,17,\ldots\}$?

Alphabets and Strings

An **alphabet** is a set of symbols

Example Alphabet: $\Sigma = \{a, b\}$

A **string** is a sequence of symbols from the alphabet

Example Strings

- a
- ab
- $abba$
- $aaabbbaabab$

String variables

- $u = ab$
- $v = bbbaaa$
- $w = abba$

Decimal numbers alphabet $\Sigma = \{0,1,2,\ldots,9\}$

102345 567463386

Binary numbers alphabet $\Sigma = \{0,1\}$

100010001 101101111

* real number alphabet $\Sigma = \{\bullet, 0,1,2,3,\ldots,9\}$

real number "32134.79201"

لازم تظهر مرة واحدة

Unary numbers alphabet $\Sigma = \{1\}$

Unary number:	1	11	111	1111	11111
Decimal number:	1	2	3	4	5

String Operations:

$w = a_1 a_2 \cdots a_n$ $\quad abba$

$v = b_1 b_2 \cdots b_m$ $\quad bbbaaa$

Concatenation

$wv = a_1 a_2 \cdots a_n b_1 b_2 \cdots b_m$ $\quad abbabbbaaa$

$vw = bbbaaaabba$



$wv \neq vw$

$\Sigma = \{a, b\}$ فرضنا

(2)

$w = a_1 a_2 \cdots a_n$ $\quad ababaaabbb$

Reverse

$w^R = a_n \cdots a_2 a_1$ $\quad bbbaaababa$

String Length

(3)

$w = a_1 a_2 \cdots a_n$

Length: $|w| = n$ or Len(w) = n or Length(w) = n

Examples:

$|abba| = 4$

$|aa| = 2$

$|a| = 1$

$|\lambda| = 0$

$|\varepsilon| = 0$

λ, ε = empty string.

Length of Concatenation

$$|uv| = |u| + |v|$$

Example: $u = aab, \quad |u| = 3$

$v = abaab, \quad |v| = 5$

$|uu| = 6$

$|uuu| = 3 * |u|$
$= 3 * 3 = 9$

انتبه هون بدجمع ما بدضرب

$$|uv| = |aababaaab| = 8$$

$$|uv| = |u| + |v| = 3 + 5 = 8$$

Empty String

A string with no letters is denoted: λ or ε

lambda — epsilon.

Acts as a neutral element

Observations: $|\lambda| = 0$

① "abc" = "λaλbλc"

② λλλ → λ

$\lambda w = w\lambda = w$

* تكرار λ نتيجته λ واحدة.

$\lambda abba = abba\lambda = ab\lambda ba = abba$

Substring

Substring of string: *a subsequence of consecutive characters*

* متتابعين

String	Substring
abbab	*ab*
abbab	*abba*
abbab	*b*
abbab	*bbab*

بتعبر عن concatenation.

"12" • "569"

"12569"

13 2ndt 22/23

13

Prefix and Suffix

string *abbab*

Prefixes	Suffixes
λ	*abbab*
a	*bbab*
ab	*bab*
abb	*ab*
abba	*b*
abbab	λ

$w = uv$

prefix: substring في البداية

suffix: substring في النهاية.

14 2ndt 22/23

14

Exponent Operation

Positive integer.

$$w^n = \underbrace{ww\cdots w}_{n}$$

Example:

$$(abba)^2 = abbaabba$$

ex:
$(011)^0 = \lambda$
$(011)^4 = 011\ 011\ 011\ 011$
$= ((011)^2)^2$

Definition: $w^0 = \lambda$

$$(abba)^0 = \lambda$$

ex:
$(hi)^0 = \lambda$
$(hi)^3 = hi\ hi\ hi$
$(hi)^1 = hi$

$N_a(w)$ – number of symbol *a* in string *w*

$$W = aabbcaaba$$

$N_a(w)=5$
$N_b(w)=3$
$N_c(w)=1$
$N_d(w)=0$

ex:
$L = \{ w : N_a(w) > N_b(w) \}$
$= \{ aba, aaba, aabbaba \cdots \}$
عدد الـ a أكبر من b في الـ string.

The * Operation

Kleene-Closure $* = 0, 1, 2, \ldots \infty$

Σ^*: the set of all possible strings from alphabet Σ

$\Sigma = \{a, b\}$

$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$

0 1 2 3

$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cdots \Sigma^\infty$

$\Sigma^* = \bigcup_{i=0}^{\infty} \Sigma^i$

$\Sigma = \{1, 2, 3\}$

"23121311 3" $\in \Sigma^*$

$\in \{1,2,3\}^*$

$\in \{1,2,3\}^9$



$\Sigma^0 = \{\lambda\}$

$\Sigma^1 = \{1, 2, 3\}$, $|\Sigma^1| = |\Sigma|^1 = 3^1 = 3$

$\Sigma^2 = \{11, 12, 13, 21, 22, 23, 31, 32, 33\}$, $|\Sigma^2| = 3^2$

$\Sigma^3 = \{111, 112, 113, 121, 122, 123, 131, 132, 133, 211, 212, 213, 221, 222, 223, 231, 232, 233, 311, 312, 313, 321, 322, 323, 331, 332, 333\}$, $|\Sigma^3| = |\Sigma|^3 = 3^3$

$\Sigma^4 = \{1111, 1112, \ldots 3333\}$ $|\Sigma^4| = |\Sigma|^4 = 3^4$

$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \ldots$

$= \{\lambda, 1, 2, 3, 11, 12, \ldots 33, 111, 112, \ldots, 333, 1111, \ldots, 3333, \ldots\}$

The + Operation (Positive Closure)

Σ^+ : the set of all possible strings from Σ alphabet except λ .

+ : 1, 2, 3 ...

$$\Sigma^+ = \bigcup_{k=1}^{\infty} \Sigma^k$$

$$= \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \dots$$

$\Sigma = \{a, b\}$

$\Sigma^* = \{\underbrace{\lambda}_{0}, \underbrace{a, b}_{1}, \underbrace{aa, ab, ba, bb}_{2}, aaa, aab, \dots\}$

* لو سأل كم عدد الـ string إن Len=3 بكون $2^3 = 8$

$\Sigma^+ = \Sigma^* - \{\lambda\}$

$$\Sigma^+ = \Sigma^* - \{\lambda\}$$

$\Sigma^+ = \{a, b, aa, ab, ba, bb, aaa, aab, \dots\}$

① $\lambda \in \Sigma^*$

② $\lambda \notin \Sigma^+$

③ $\Sigma^* \cap \Sigma^+ = \Sigma^+$

④ $\Sigma^* \cup \Sigma^+ = \Sigma^*$

⑤ $\Sigma^* - \{\lambda\} = \Sigma^+$

19

$$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \dots$$

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \dots$$

λ

Languages

A language over alphabet Σ is any subset of Σ^*

universal

Example:

$\Sigma = \{a, b\}$

$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, \dots\}$ (aab)

Finite languages

Language1: $\{\lambda\}$ — $|\text{language1}| = 1$

Language2: $\{a, aa, aab\}$ — $|\text{language2}| = 3$

Language3: $\{\lambda, abba, baba, aa, ab, aaaaaa\}$ — $|\text{language3}| = 6$

20

More Language Examples

Alphabet $\Sigma = \{a, b\}$

An infinite language $L = \{a^n b^n : n \geq 0\}$

L-R

$a^0 b^0 = \lambda\lambda = \lambda$

$a^1 b^1 = ab$

$a^2 b^2 = aabb$

$a^5 b^5 = aaaaa\,bbbbb$

- λ (n = 0)
- ab (n = 1)
- $aabb$ (n = 2)
- $aaaaabbbbb$ (n = 5)

$\in L$

لا يحقق الشرط: عدد a نفس عدد b

- $bbabb \notin L$ ($\in \Sigma^*$)
- $abb \notin L$ ($\in \Sigma^*$)

More Language Examples

$\Sigma = \{a, b, c, d, \ldots z\}$

$\Sigma^* = \{\lambda, a, b, c \ldots z, aa, \ldots ar, \ldots hi, \ldots to, \ldots of, \ldots aaa, aab, \ldots the, \ldots sun, \ldots aaaa, \ldots$ "home", ... 'list', ...

$L_{eng} \subset \Sigma^*$

* String operation

① Na(w) ...
2) Length .... |w|
③ concat.... u.w
4) Reverse .... w^R
5) exponent .... w^n ⇒ $n \geq 0$
6) Kleene closure * operation
7) positive closure +

3/1/202

More Language Examples

$\Sigma = \{0, 1\}$

① $L = \{w : w \in \Sigma^*\}$
$L = \{\lambda, 0, 1, 00, 01, 10, 11, \ldots\}$

2) $L = \{0w : w \in \Sigma^*\}$
$L = \{0, 00, 01, 000, 001, \ldots\}$

3) $L = \{w1 : w \in \Sigma^*\}$
$L = \{1, 01, 11, 001, 011, 101, 111, \ldots\}$

4) $L = \{00w11 : w \in \Sigma^+\}$
$L = \{0011, 00111, 001011, \ldots\}$

5) $L = \{1^k : k \geq 0\}$
$L = \{\lambda, 1, 11, 111, \ldots\}$

6) $L = \{(01)^m : m \geq 2\}$ ⇒ $m = 2, 3, 4, 5, \ldots$
$L = \{0101, 010101, \ldots\}$
$(01)^2 = 0101$
$(01)^3 = 010101$

7) $L = \{ww^R : w \in \{0,1\}^*\}$
$\lambda\lambda^R = \lambda$
$L = \{\lambda, 00, 11, 0110, \ldots\}$
$01(01)^R = 0110$

$\{0,1\}^* = \{\lambda, 0, 1, 01, \ldots\}$

8) $L = \{u : u \in \Sigma^* \text{ and } 2 \leq |u| \leq 3\}$
$L = \{0, 1, 00, 01, 10, 11, \ldots, 111\}$
$|L| = 2 + 4 + 8 = 14$
$2^1, 2^2, 2^3$

λ ما راح تكون من ضمن الـ Language لانو اقل len = 1

23

Prime numbers

Alphabet $\Sigma = \{0,1,2,\ldots,9\}$

Language:

$$PRIMES = \{x : x \in \Sigma^* \text{ and } x \text{ is prime}\}$$

$$PRIMES = \{2,3,5,7,11,13,17,\ldots\}$$

24

Even and odd numbers

Alphabet $\Sigma = \{0,1,2,\ldots,9\}$

Languages: $EVEN = \{x : x \in \Sigma^* \text{ and } x \text{ is even}\}$

$EVEN = \{0,2,4,6,\ldots\}$

$ODD = \{x : x \in \Sigma^* \text{ and } x \text{ is odd}\}$

$ODD = \{1,3,5,7,\ldots\}$

$L_{odd} \subseteq \Sigma^* \equiv L_{odd} \subseteq \{0,1,2,\ldots,9\}^*$

$L_{even} \cup L_{odd} = \{0,1,2,\ldots,9\}^*$

$L_{even} \cap L_{prime} = \emptyset$ $\quad L_{prime} \subseteq L_{odd}$

$L_{odd} \cap L_{prime} = L_{prime}$

25 2ndt 22/23

25

Addition (of unary numbers)

Alphabet: $\Sigma = \{1,+,=\}$

Language: $ADDITION = \{x + y = z : x = 1^n, y = 1^m, z = 1^k, n + m = k\}$

$11 + 111 = 11111 \in ADDITION$

$111 + 111 = 111 \notin ADDITION$

26 2ndt 22/23

26

Squares (of unary numbers)

Alphabet: $\Sigma = \{1, \#\}$

Language: $SQUARES = \{x\#y : x = 1^n, y = 1^m, m = n^2\}$ $n \geq 1$ $m \geq 1$

$11\#1111 \in SQUARES$

$111\#1111 \notin SQUARES$

؟

اسأل كيف m معتمده على n وهو لا يقرأ.

27 2ndt 22/23

27

Two special languages

1) Empty language

$\{\} = \emptyset$

2) Language with empty string

$\{\lambda\}$

$\{\}^* = \{\lambda\}$

$\overset{\text{size}}{\overbrace{\{\}}} \neq \overset{\text{size}=1}{\overbrace{\{\lambda\}}}$

Size of a language (number of elements):

$|\{\}| = 0$

$|\{\lambda\}| = 1$

$|\{a, aa, ab\}| = 3$

$|\{\lambda, aa, bb, abba, baba\}| = 5$

28 2ndt 22/23

28

Note that:

Sets $\emptyset = \{\} \neq \{\lambda\}$

Set size $|\{\}| = |\emptyset| = 0$

Set size $|\{\lambda\}| = 1$

String length $|\lambda| = 0$

$w^0 = \lambda$

$\{\}^* = \{\lambda\}$

$\{\}^0 = \{\lambda\}$

$\{\}^1 = \{\}$ لا يوجد شيء لتكراره، كما لا يوماني الشيء يكون اذا $len = 0$

Operations on Languages

The usual set operations

* لا تكرر لانها set ← لا يجوز التكرار

1. $\{a, ab, aaaa\} \cup \{bb, ab\} = \{a, ab, bb, aaaa\}$
2. $\{a, ab, aaaa\} \cap \{bb, ab\} = \{ab\}$
3. $\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$
4. **Complement:** $\bar{L} = \Sigma^* - L$ ضمن دائرة

$\overline{\{a, ba\}} = \{\lambda, b, aa, ab, bb, aaa, \ldots\}$

* $L_1 = \{aa, ab\}$ عرة $L_2 = \{ba, ab\}$ عرة

L_1 XoR $L_2 = \{aa, ba\}$

5.Reverse

Definition: $L^R = \{w^R : w \in L\}$

Examples:

$\{ab, aab, baba\}^R = \{ba, baa, abab\}$

$L = \{a^n b^n : n \geq 0\}$, $L^R = \{b^n a^n : n \geq 0\}$

$\{\lambda, bba, aaabb\}^R = \{\lambda, abb, bbaaa\}$

$L = \{a w_1 b w_2 : w_1 \in \{0,1\}^*, w_2 \in \{a,b\}^*\}$

$L^R = \{w_1 b w_2 a : w_1 \in \{a,b\}^*, w_2 \in \{0,1\}^*\}$

31 ملاحظة: يقدر اغير اسمهم زي ال w2 غيرتهاد اس و ال اس لـ w2 مع تغير الشروط.

6.Concatenation

Definition:

$L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$

Example:

$\{a, ab, ba\}\{b, aa\} = \{ab, aaa, abb, abaa, bab, baaa\}$

$|L_1 L_2| = 3 * 2 = 6$ لا مش شرط

$\{\lambda, aa\}\{\lambda, aa, bb\} = \{\lambda, aa, bb, aaaa, aabb\}$

$|L_1 L_2| = 5$

aa, لم نكرر

كيف ظهروا بس 5 عناصر؟

$\lambda \cdot aa = aa$
$aa \cdot \lambda = aa$ → نوعين وحدة

$|L_1 L_2| \leq 6$ ← وإذا بكون

32 بغير الرمز

$L_1 = \{a^n : n \geq 1\}$, $L_2 = \{b^n : n \geq 0\}$

$L_1 L_2 = \{a^m b^k : m \geq 1, k \geq 0\}$

$L_1 L_1 = \{a^i a^j : i \geq 1, j \geq 1\}$

Another Operation- 7.exponent operation

$str^0 = \lambda$

$\Sigma^0 = \{\lambda\}$

Definition: $n \geq 0$

$$L^n = \underbrace{LL\cdots L}_{n}$$

Example: $\{a,b\}^3 = \{a,b\}\{a,b\}\{a,b\} = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$

$\{01, 11\}^2 = \{01, 11\}\cdot\{01, 11\} = \{0101, 0111, 1101, 1111\}$

special case: $L^0 = \{\lambda\}$

$\{a, bba, aaa\}^0 = \{\lambda\}$

Example $L = \{a^n b^n : n \geq 0\}$

$L^2 = \{a^n b^n a^m b^m : n, m \geq 0\}$

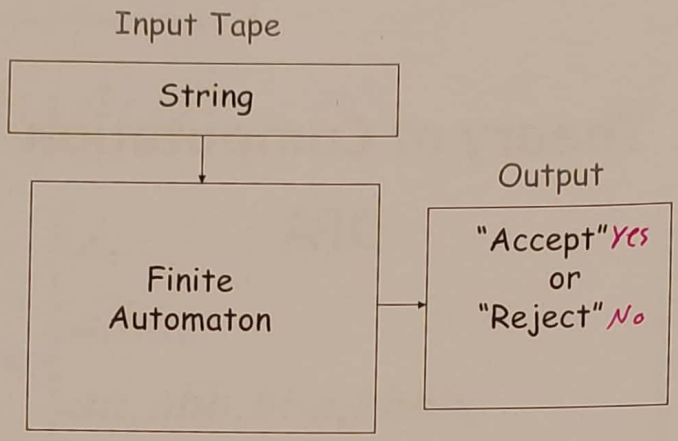
$aabbaaabbb \in L^2$

Theory of Computation
DFA

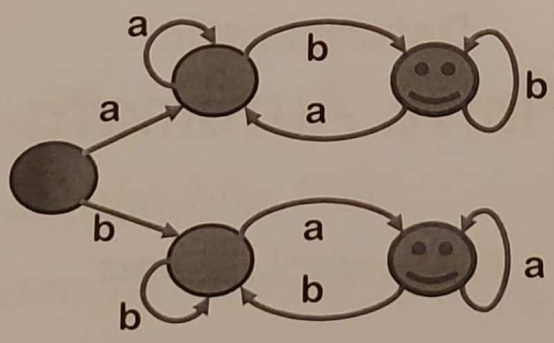
Deterministic
Finite Automata

And Regular Languages

Deterministic Finite Automaton (DFA)



Finite Automata



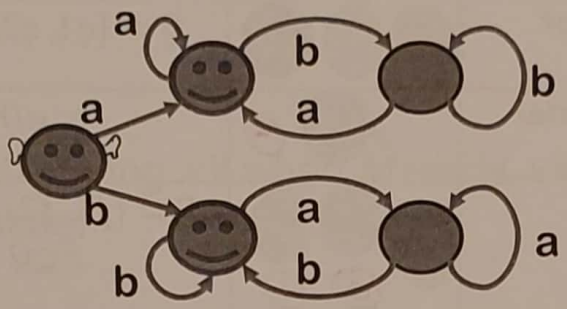
end state ترمز الى ← ◎

start state. ترمز إلى ← →○

(2)

Meet "ABA" The Automaton!

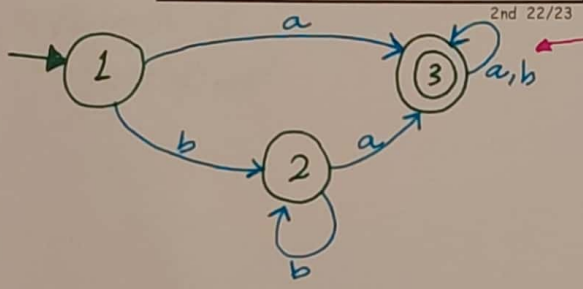
(-1) ترمز انها start state.



Input String	Result
aba	Accept
aabb	Reject
aabba	Accept
ε or λ	Accept

δ x	a	b
-1	3	2
2	3	2
+3	3	3

Q ↕ — Σ ↔



ex:

* $Q = \{1, 2, 3\}$ اذا ← Q * عندي 3 state

* $\Sigma = \{a, b\}$ * $q_0 = 1$

* $F = \{3\}$

* $\delta(1, a) \to 3$

$\delta(1, b) \to 2$

$\delta(2, a) \to 3$

$\delta(2, b) \to 2$

$\delta(3, a) \to 3$

$\delta(3, b) \to 3$

اول اشي

Formal Definition

Deterministic Finite Automaton (DFA)

احفظ الترتيب احتياط ← $M = (Q, \Sigma, \delta, q_0, F)$

Q : set of states

Σ : input alphabet $\lambda \notin \Sigma$

δ : transition function → $\delta : Q \times \Sigma \to Q$

بتكتب اذ Q

بعدين ع

بعض (b,3) غلط

start state ← q_0 : initial state

$q_0 \in Q$

F : set of accepting states

$F \subseteq Q$

* دوالر

Definition DFA

Q	Finite set of states	$Q = \{q1, q2, \ldots qn\}$
q_0	A start state/ initial state	$q_0 \in Q$
F	A set of accepting states	$F = \{f1, f2, \ldots . fm\}$ $F \subseteq Q$
Σ	A finite alphabet	a b # x 1 — $\sum$ $\lambda \notin \Sigma$
δ	State transition instructions	$\partial : Q \times \Sigma \to Q$ $\partial(q_i, a) = q_j$

اعرف الكلام

How Machine M **operates.**

M "reads" **one** letter at a time from the input string(Left to Right)

M starts in state q_0.

If M is in state q_i reads the letter *a* then
- If $\delta(q_i, a)$ is undefined then
  - CRASH.
- Otherwise
  - M **moves** to state $\delta(q_i, a)$

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a finite automaton.

1. *M* ***accepts*** the string *x* if when *M* reads *x* it ends in an **accepting** state.
2. *M* ***rejects*** the string *x* if when *M* reads *x* it ends in a non-accepting state.
3. *M* ***crashes*** on *x* if *M* crashes while reading *x*.

إذا كان احد الامرين
في الـ string x
undefined.

To accept a string:

all the input string is scanned and the last state is accepting

To reject a string:

all the input string is scanned and the last state is non-accepting (crash)

The set (or language) accepted by M is:

$$L_M = \{x \in \Sigma^* \mid M \text{ accepts } x\}$$

$\Sigma^k \equiv$ All length k strings over the alphabet Σ

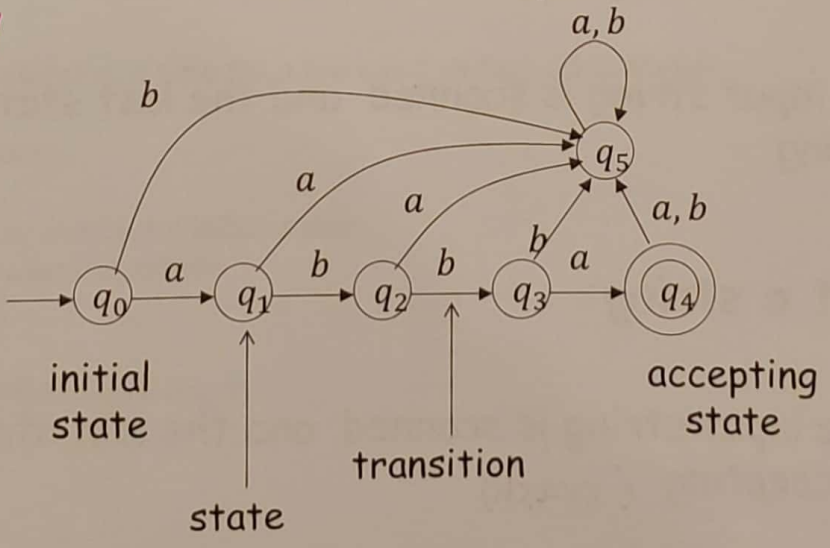
$\Sigma^* \equiv \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots$

Notice that this is $\{\varepsilon\}$ or $\{\lambda\}$

Transition عدد ال *

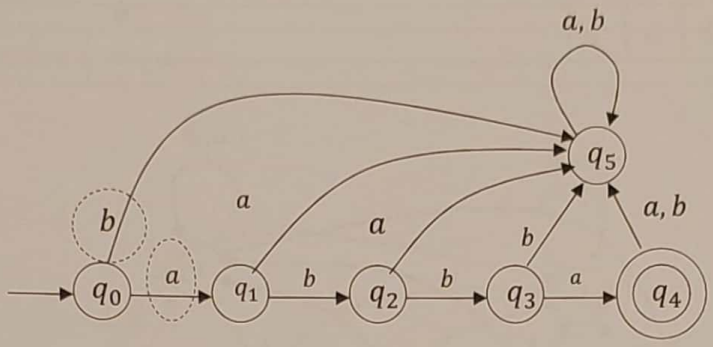
$|\delta| = |Q| \cdot |\Sigma|$
$= 6 \cdot 2$
$= 12$

Transition Graph



- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{a, b\}$ - δ --
- $q_0 = q_0$ - $F = \{q_4\}$

Alphabet $\Sigma = \{a, b\}$



For every state, there is a transition
for every symbol in the alphabet

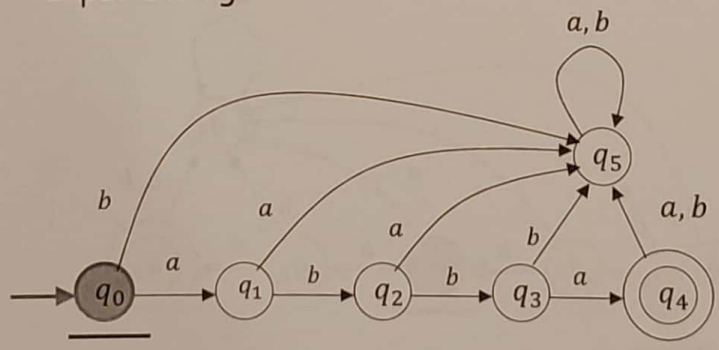
Initial Configuration

head

Input Tape

a	b	b	a	

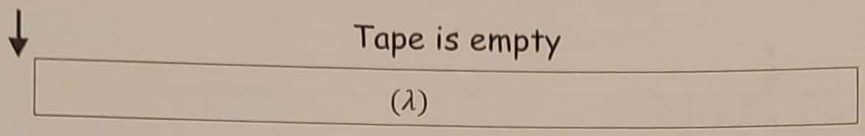
Input String



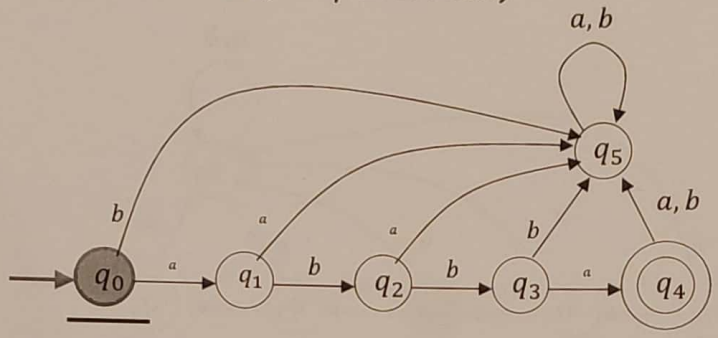
Initial state

7

Another Rejection Case



Input Finished (no symbol read)



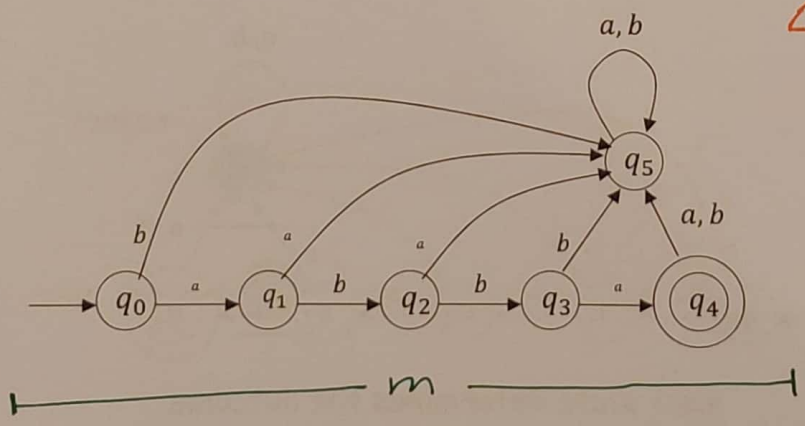
reject

This automaton accepts only one string

$L(m) = \{abba\}$ $\Sigma = \{a, b\}$

$L(m) \subseteq \Sigma^*$

Language Accepted: $L = \{abba\}$



test

λ → Reject

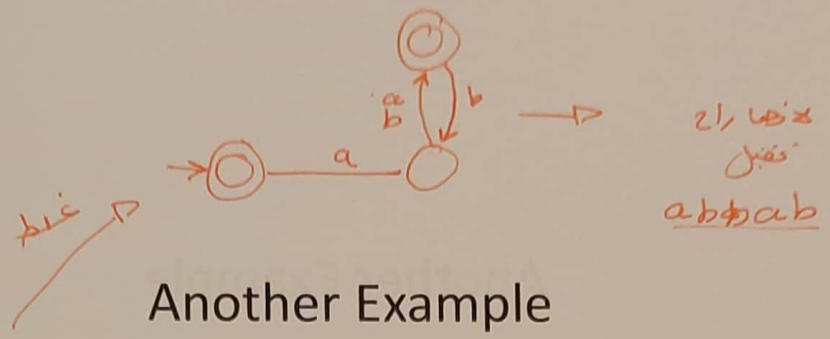
a → Reject

b → Reject

aa → Reject

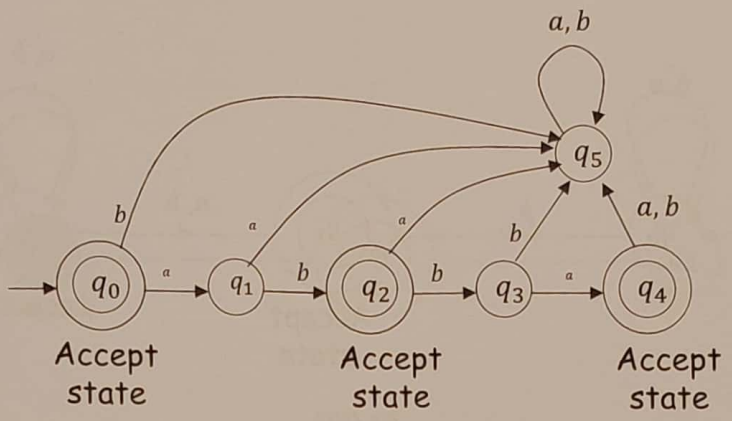
⋮

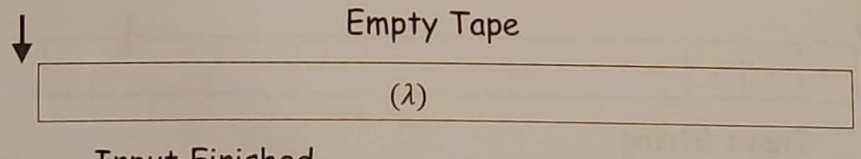
abba → accep



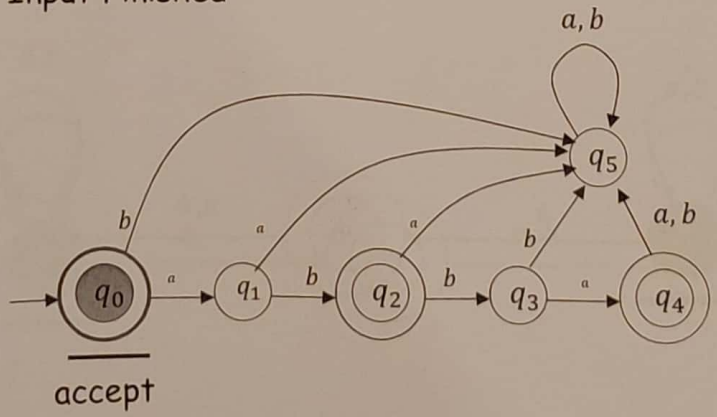
Another Example

$L = \{\lambda, ab, abba\}$



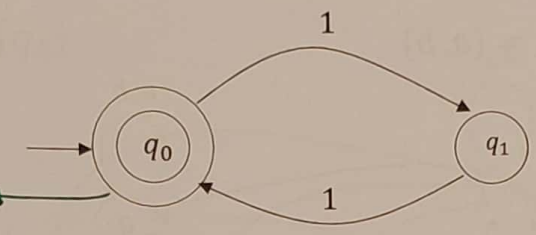


Input Finished



Another Example

Alphabet: $\Sigma = \{1\}$



start الوقت بنفس end state.

Language Accepted:

$$EVEN = \{x : x \in \Sigma^* \text{ and } x \text{ is even}\}$$
$$= \{\lambda,\ 11,\ 1111,\ 111111,\ \ldots\}$$

$L = \{1^k \mid k \text{ is even}\}$??

* $Q = \{q_0, q_1\}$
* $\Sigma = \{1\}$
* $\delta =$

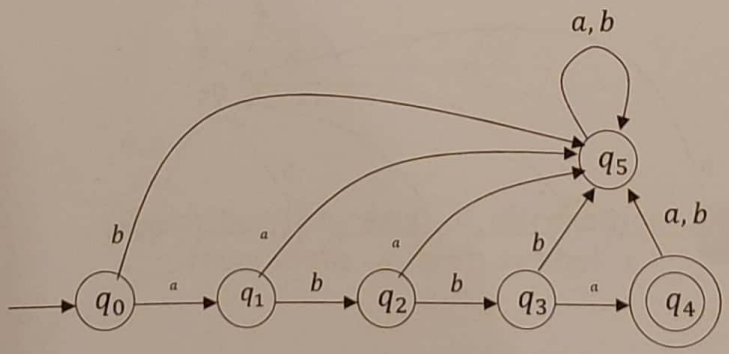
	1
q_0	q_1
q_1	q_0

* $q_0 = q_0$
* $F = \{q_0\}$

Set of States Q

Example

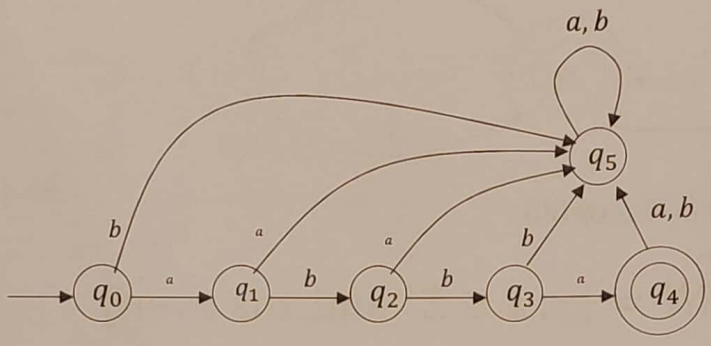
$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$



Input Alphabet Σ

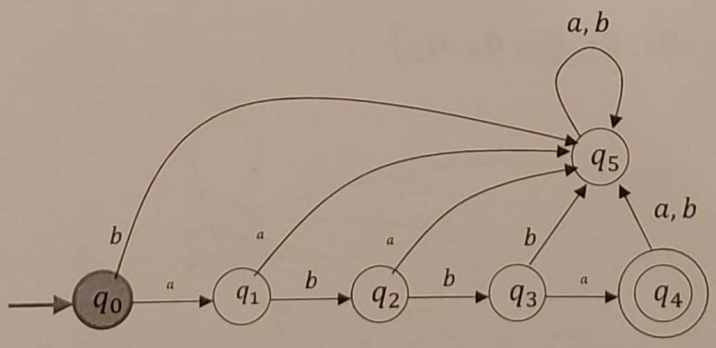
Note: the input alphabet never contains λ $\lambda \notin \Sigma$

Example $\Sigma = \{a, b\}$



Initial State q_0

Example



* الـ Path حالة خاصة من الـ walk
حالة اعم

Extended Transition Function

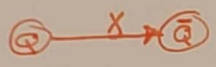
Path/walk يعبر عن $\delta^*: Q \times \Sigma^* \to Q$

$$\delta^*(q, w) = q'$$

String

$w \in \Sigma^*$ ← مش لـ Σ

Describes the resulting state after scanning string w from state q



extended transition function.

* الفرق بين δ و δ^*

47

* δ بتعبر عن انو انا موجود في q وبدي انتقل لـ $\overline{q}$ عن طريق X بكتبها بطريقة التالية

$\delta = (q, X) \to \overline{q}$

Example: حرف

* δ^* إذا كنت بـ state معينة وعندي string مكون مثلاً من $a_1 a_2 a_3 \dots$ حتى توصل لـ end state

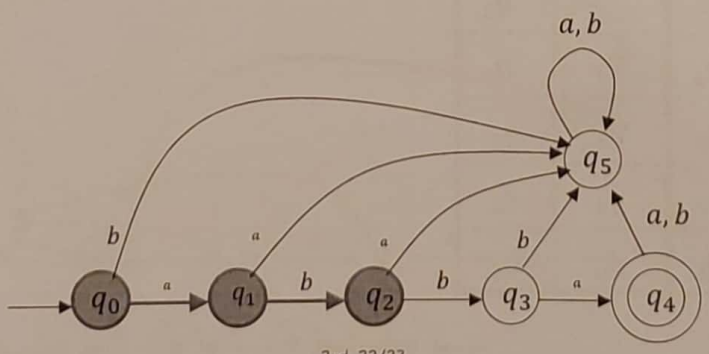
* بدي استخدم كل الحروف اللي بالـ string

$\delta^*(q, w) = \overline{q}$

String

$\delta^*(q_0, ab) = q_2$

$q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_2$



48

δ بكون
Q × Σ
وΣ ما بتحتوي
على λ وبرضو
λ تعتبر
string

لازم

لانو

غلط $\delta(q,\lambda)=q_1$ إذا

Special case:

ليش مش q؟

$\delta(q,\lambda)=\bar{q}$

δ والا لا تقبل
إلا string

for any state q

$$\delta^*(q,\lambda)=q$$

* يعني إذا قرأت λ بظل
في نفس الـ state

ex:

$\delta^*(1,\lambda)=1 \rightsquigarrow$ No $1\notin F$

$\delta^*(1)\equiv\delta^*(1,\lambda)=1 \rightsquigarrow$ No

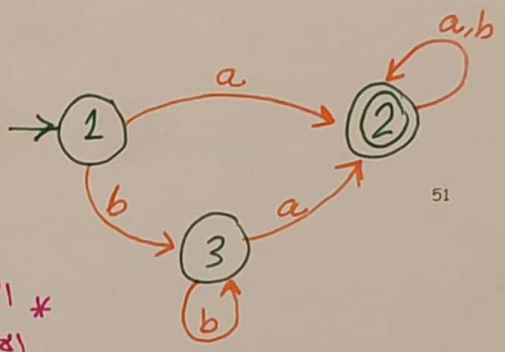
$\delta^*(3,aba)=2$

$\delta^*(3,bb)=3$ } ما بقدر أحدد

$\delta^*(1,ba)=2 \rightsquigarrow$ yes

$\delta^*(1,ab)=2 \rightsquigarrow$ yes

$\delta^*(1,bb)=3 \rightsquigarrow$ No



51

2nd 22/23

51

بلشو من 3 مش من 1

1 start state.

* الحالات اللي بلون
الأخضر ما بقدر أحدد أنهم
accept أو Reject
لأنهم ما بلشو من الـ start state

In general:

$$\delta^*(q,w)=q'$$

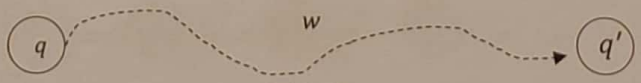
implies that there is a **walk** of transitions

$$w=\sigma_1\sigma_2\cdots\sigma_k$$

$w\in\Sigma^*$



states may be **repeated**



2nd 22/23

52

52

Language Accepted by DFA
$L(M)$ or L_M

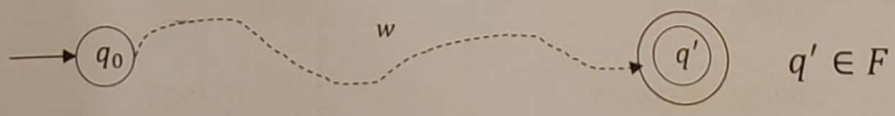
Language accepted by DFA M:

- it is denoted as $L(M)$ and contains all the strings accepted by M
- We also say that M recognizes $L(M)$

For a DFA $M = (Q, \Sigma, \delta, q_0, F)$

Language accepted by M :

$$L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$$



Language rejected by M :

$$\overline{L(M)} = \{w \in \Sigma^* : \delta^*(q_0, w) \notin F\}$$



non-accepting state.

More DFA Examples

$\Sigma = \{a, b\}$

* ممكن تكون m تتكون من state واحدة

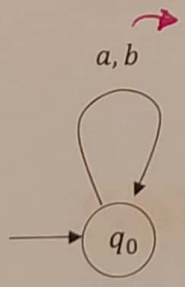
$Q = \{q_0\}$

$\Sigma = \{a, b\}$

$\delta = \delta(q_0, a) \to q_0$
$\delta(q_0, b) \to q_0$

$q_0 = q_0$

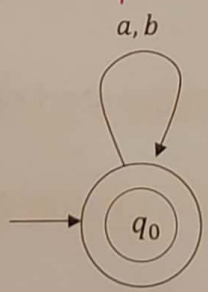
$F = \{\}$



ما بتعمل accept لاي شي

$L(M) = \{\}$

Empty language



accept بتعمل لاي شي

$L(M) = \Sigma^*$

All strings

شو ما ادخل راح يعطيني accept.

$Q = \{q_0\}$

$F = \{q_0\}$

مثلاً دخلت q راح يعطيني accept ليش؟ لانو راح يضل في q_0 و q_0 هي ال accepting state.

ممكن هيك اكتبها . $Q=\{0,1\}$ ← $Q=\{q_0, q_1\}$

$\Sigma=\{a,b\}$

$\delta=$

	a	b
$\to$ (+) 0	1	1
1	1	1

end , start نفسها state.

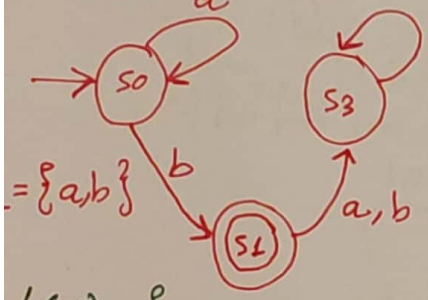
$\Sigma=\{a,b\}$ إذا طلب جدول بالاستثناء لازم نكتبهم

$q_0=0$

$F=\{1\}$

ex:

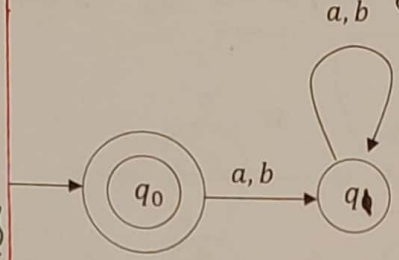
H:



$\Sigma=\{a,b\}$

$L(H)=\{b, ab, aab, aaab, \ldots\}$

$L(H)=\{a^k b : k \geq 0\}$



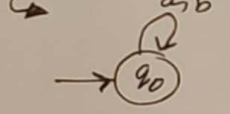
$L(M)=\{\lambda\}$

الوحيد الي بعمل accept

Language of the empty string

تختلف عن

$L(M)=\{\}$



$\Sigma=\{a,b\}$

2nd 22/23

57

لانو

$\delta^*(0,\lambda)\to 0$ ~> Yes

$\delta^*(0,a)\to 1$ ~> No

$\delta^*(0,b)\to 1$ ~> No

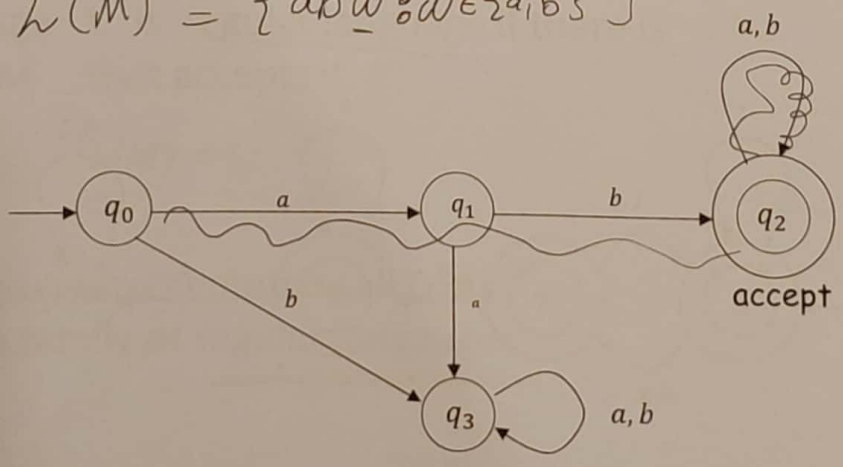
$\delta^*(0,abab)\to 1$ ~> No

57

$L(M)$ = { all strings with prefix ab }

$L(M)=\{abw : w\in\{a,b\}^*\}$

M:



2nd 22/23

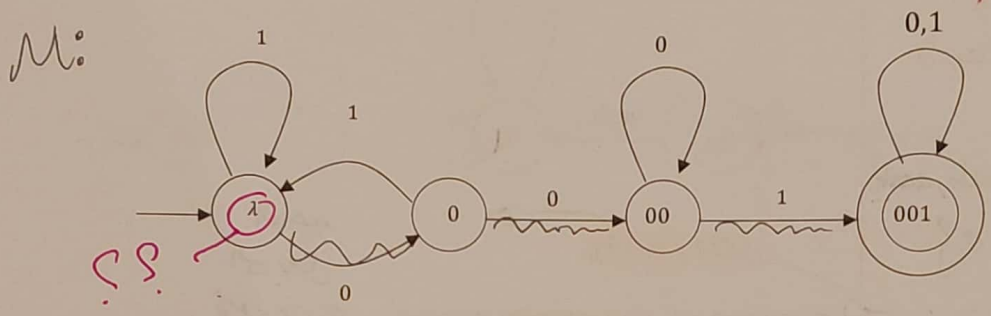
58

58

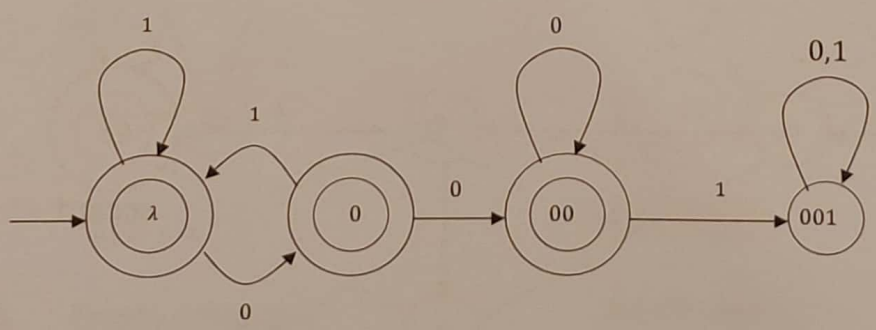
$L(M)$ = { all binary strings containing substring 001 }

$L(M) = \{ w_1 001 w_2 : w_1 \in \{0,1\}^* \text{ and } w_2 \in \{0,1\}^* \}$

M:



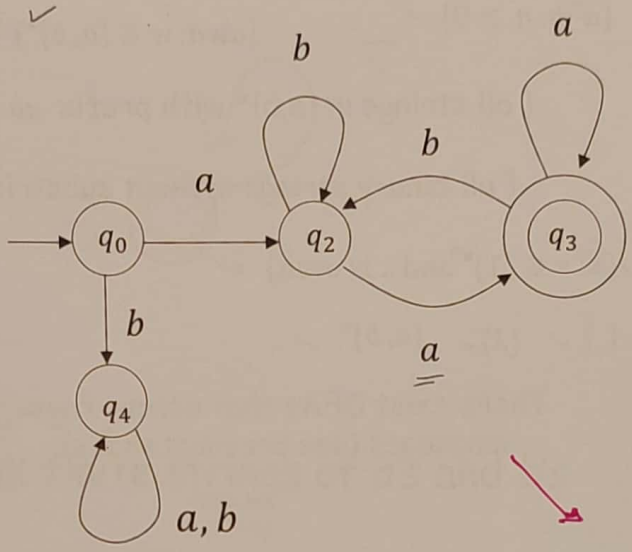
L(M)= { all binary strings without substring **001** }



$$L(M) = \{awa : w \in \{a,b\}^*\}$$

aabbbbaaba ✓

baaaba ✗





Regular Languages

Definition:

A language L is regular language if there is a DFA M that accepts it.

$$L(M) = L$$

The languages accepted by all DFAs form the family of regular languages

Example regular languages:

$\{abba\}$ ✓ $\{\lambda, ab, abba\}$ ✓

$\{a^n b : n \geq 0\}$ ✓ $\{awa : w \in \{a,b\}^*\}$ ✓

{ all strings in {a,b}* with prefix ab } ✓

{ all binary strings without substring 001 } ✓

$\{x : x \in \{1\}^* \text{ and } x \text{ is even}\}$ ✓

$\{\}$ ✓ $\{\lambda\}$ ✓ $\{a,b\}^*$ ✓

There exist DFAs that accept these languages (see previous slides).

There exist languages which are not Regular:

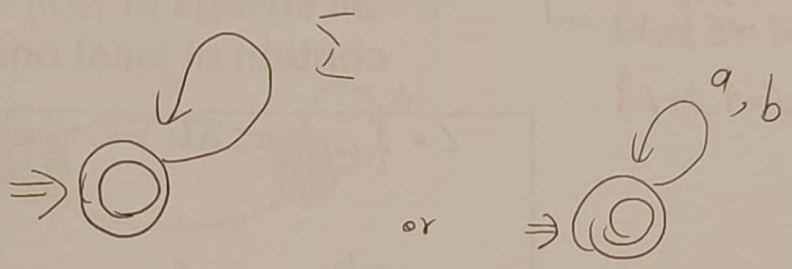
✗ $L = \{a^n b^n : n \geq 0\}$

$$ADDITION = \{x + y = z \ : \mathrm{x} = 1^{\mathrm{n}}, \mathrm{y} = 1^{\mathrm{m}}, \mathrm{z} = 1^{k}, \ \mathrm{n} + m = k\}$$

There are no DFAs that accept these languages

(we will prove this in a later class)

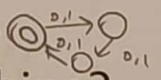
Design DFA to accept L?



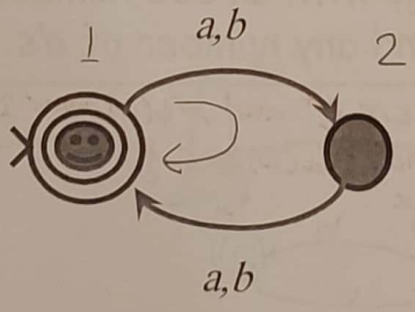
$L = \{a,b\}^* = \Sigma^*$

=all finite strings of *a*'s and *b*'s

$L(M) = \{w : w \in \{0,1\}^* \text{ and } |w| \bmod 3 = 0\}$



What is the language accepted by this machine?



$\Sigma = \{a, b\}$

$0 = |w|$ λ ✓

$2 = |w|$ aa ✓

$2 = |w|$ ab ✓

ba ✓

bb ✗

$4 = |w|$ aaaa ✓

$6 = |w|$ aaaaaa

$L = \{w : w \in \{a,b\}^* \text{ and } |w| \bmod 2 = 0\}$

$L = \{w : w \in \{a,b\}^* \text{ and } |w| \text{ is even}\}$

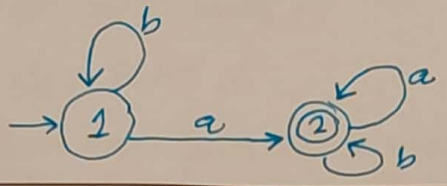
What machine accepts this language?

L = **all strings in** $\{a,b\}^*$ **that contain at least one** a

$L = \{ w \mid w \in \{a,b\}^* \text{ and } N_a(w) \geq 1 \}$

$N_a(w) \geq 1$

$L = \{ a, aa, ab, ba, \ldots, bbbabbbb \}$ $N_b(w) \geq 0$

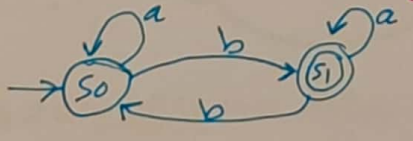


What machine accepts this language?

$b = \{1, 3, 5, 7, \ldots\}$

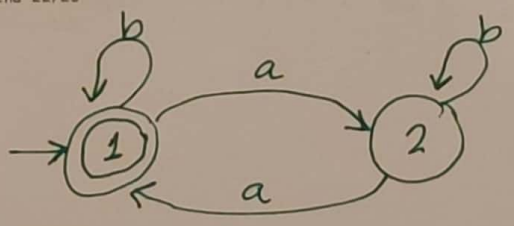
L = strings with an odd number of b's and any number of a's

$L = \{ w : w \in \Sigma^* \text{ and } N_b(w) \bmod 2 = 1 \text{ and } N_a(w) \geq 0 \}$

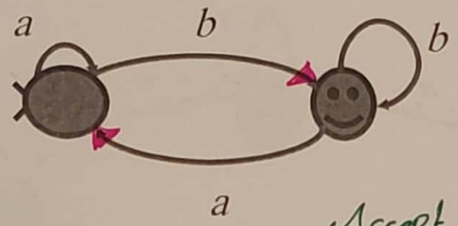


$\delta^*(q_0, \text{"abaabaabaaaabb"}) \rightarrow S_1 \in F$ ∴ Accepted

ex: $L = \{ w \in \Sigma^* \mid N_a(w) \text{ is even}, N_b(w) \geq 0 \}$



What is the language accepted by this machine?



Accept بكون ب انتهى إذا

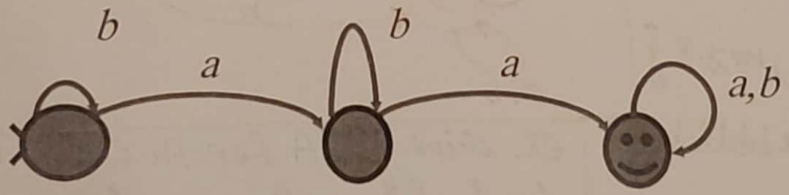
$L = \{ b, ab, aab, \dots, bb, bbbb, \dots, bab, abaaab, \dots \}$

69

$L = \{ w.b : w \in \Sigma^* \}$

عشان يكون منتهي ب b

What is the language accepted by this machine?



$L = \{ w : w \in \Sigma^* \text{ and } N_a(w) \geq 2 \}$

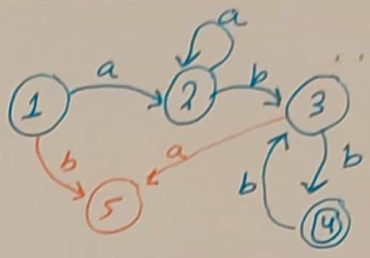
70

ex:

$L = \{ a^n b^{2m} : n \geq 1, m \geq 1 \}$ $\Sigma = \{a, b\}$

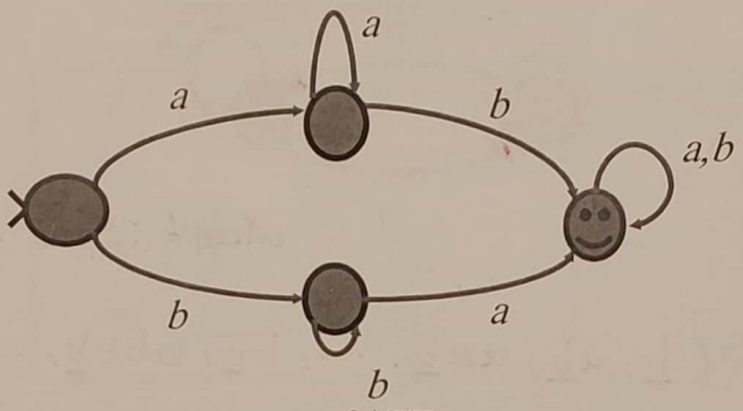
$L = \{$ abb, abbbbb, aabb, aabbbb $\}$

n=1 m=1 | n=1 m=2 | n=2 m=1 | n=2 m=2



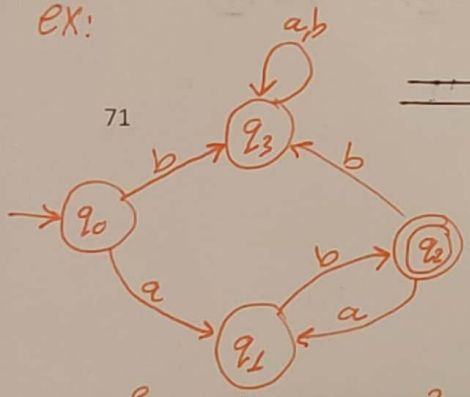
What machine accepts this language?

L = any string with an a and a b



2nd 22/23 71

ex:

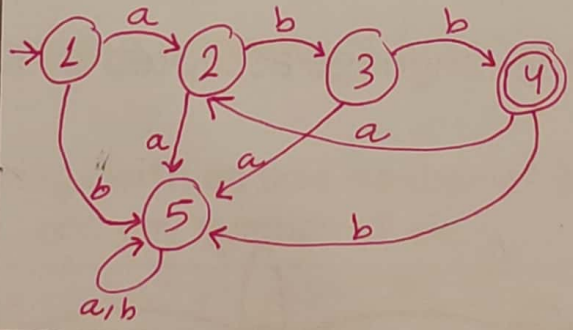


$L(m) = \{ (ab)^n : n \geq 1 \}$

ex: Design DFA for the following language.

① $L = \{ (abb)^n : n \geq 1 \}$

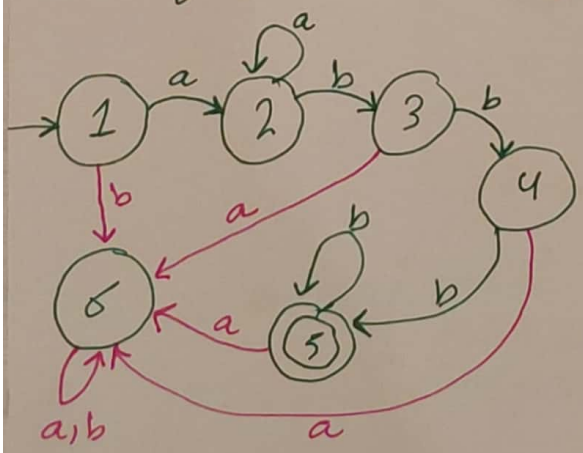
↳ $L = \{$ abb, abbabb, abbabbabb, ... $\}$



ex: Design...

$L = \{ a^n b^m : n \geq 1, m \geq 3 \}$

↳ $L = \{$ abbb, aabbbbb, ... $\}$

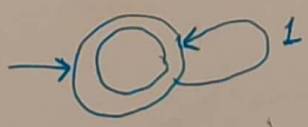


ex: Give DFA for the following language.

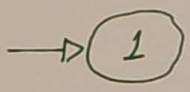
$L = \{1\}^* = \Sigma^*$, $\Sigma = \{1\}$

$L = \{ w : w \in \Sigma^*, \Sigma = \{1\} \}$

$L = \{ \lambda, 1, 11, 111, 1111, \ldots \}$



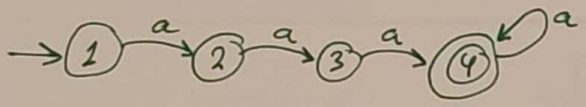
ex: Design M for the following Language.

1) $L = \phi$ 

Accept فيش فيها *

2) $L = \{\lambda, a\}$ 

بعمل Accept فيها للحالتين λ , a

3) $L = \{a^n : n \geq 3\}$ 

DFA ⊆ NFA

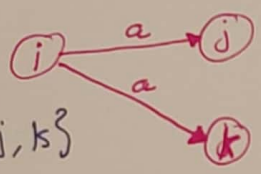
Non-Deterministic Finite Automata- NFA

Theory of Computation

3 cases

1

① $\delta(i,a)=\{j,k\}$



② $\Sigma=\{a,b\}$



↳ a is not defined

$\delta(i,a)=\emptyset$

$\delta(i,a)=\{\}$

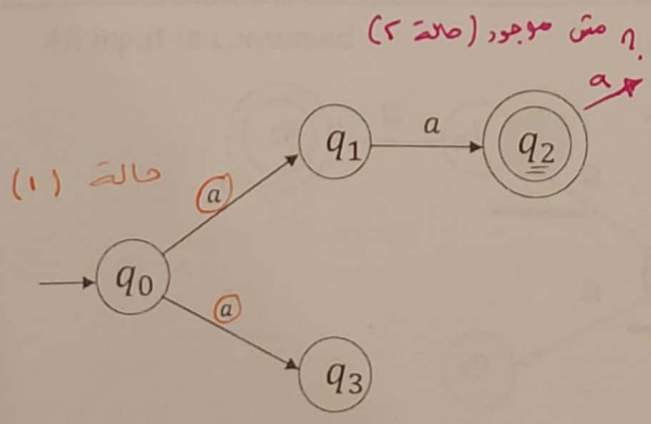
NONDETERMINISTIC FINITE AUTOMATON (NFA)

Alphabet = {a}

aa ✓

a ✗

aaa ✗



"NFA"

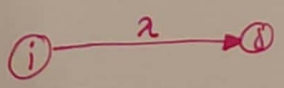
$\delta(q_0,a)=\{q_1,q_3\}$

$\delta(q_1,a)=\emptyset$

$\delta(q_3,a)=\emptyset$

2

③



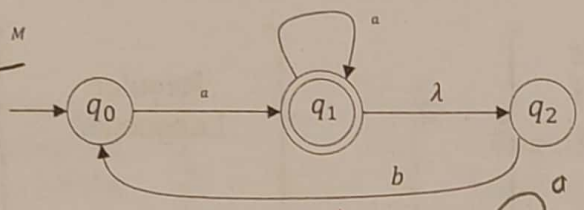
$\delta(i,\lambda)=\{j\}$

NFA في q_1 في جدول accept state الـ

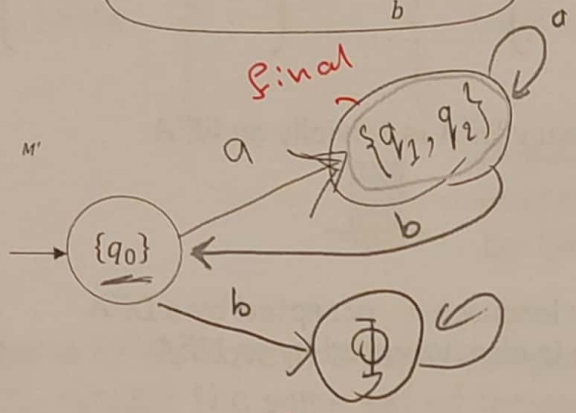
DFA في state اي q_1 تحتوي على تكون Accept state.

Conversion NFA to DFA

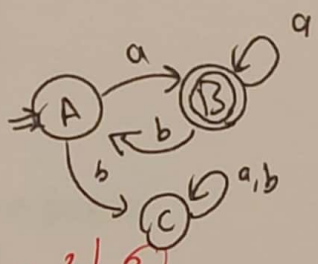
NFA M



DFA M'



- $A = \{q_0\}$
- $B = \{q_1, q_0\}$
- $C = \Phi$



75

① $\delta^*(q_0, \lambda) = \{q_0\}$ ✓ start state of DFA

② $\delta^*(\{q_0\}, a) = \{q_1, q_2\}$ 1st 22-23

③ $\delta^*(\{q_0\}, b) = \Phi$

④ $\delta^*(\{q_1, q_2\}, b) = \{q_0\}$

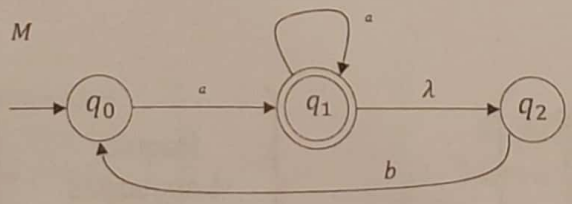
⑤ $\delta^*(\Phi, a) = \Phi$

⑥ $\delta^*(\Phi, b) = \Phi$

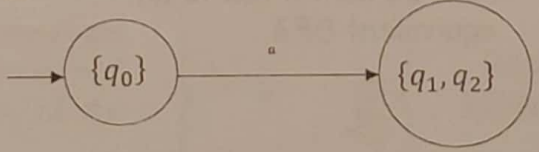
⑦ $\delta^*(\{q_1, q_2\}, a) = \{q_1, q_2\}$ ← loop

$\delta^*(q_0, a) = \{q_1, q_2\}$

NFA M



DFA M'



✓

1st 22-23

76